

$$2.4 \quad (a) \quad S \rightarrow R1R1R1R \\ R \rightarrow 0R \mid 1R \mid \epsilon$$

$$(d) \quad S \rightarrow 0 \mid 0S0 \mid 0S1 \mid 1S0 \mid 1S1$$

$$2.6 \quad (a) \quad S \rightarrow TaT$$

$$T \rightarrow TT \mid aTb \mid bTa \mid a \mid \epsilon$$

T generates all strings with at least as many a's as b's, and S forces an extra a.

$$(c) \quad S \rightarrow TX$$

$$T \rightarrow 0T0 \mid 1T1 \mid \#X$$

$$X \rightarrow 0X \mid 1X \mid \epsilon$$

2.7 (a) The PDA uses its stack to count the number of a's minus the number of b's. It enters an accepting state whenever this count is positive. In more detail, it operates as follows. The PDA scans across the input. If it sees a b and its top stack symbol is an a, it pops the stack. Similarly, if it scans an a and its top stack symbol is a b, it pops the stack. In all other cases, it pushes the input symbol onto the stack. After the PDA finishes the input, if a is on top of the stack, it accepts. Otherwise it rejects.

(c) The PDA scans across the input string and pushes every symbol it reads until it reads a #. If a # is never encountered, it rejects. Then, the PDA skips over part of the input, nondeterministically deciding when to stop skipping. At that point, it compares the next input symbols with the symbols it pops off the stack. At any disagreement, or if the input finishes while the stack is nonempty, this branch of the computation rejects. If the stack becomes empty, the machine reads the rest of the input and accepts.

$$3.2 \quad (a) \quad q_1 11, xq_3 1, x1q_3 \sqcup, x1 \sqcup q_{\text{reject}}.$$

3.8 (a) "On input string w :

1. Scan the tape and mark the first 0 that has not been marked. If no unmarked 0 is found, go to stage 4. Otherwise, move the head back to the front of the tape.
2. Scan the tape and mark the first 1 that has not been marked. If no unmarked 1 is found, *reject*.
3. Move the head back to the front of the tape and go to stage 1.
4. Move the head back to the front of the tape. Scan the tape to see if any unmarked 1s remain. If none are found, *accept*; otherwise, *reject*."